Function notation

1 The point $(7, -6)$ satisfies the function $f(x)$. Write this using function notation.

2 If $f(x) = 9x^2 + 7x - 4$, find:

a $f(-4)$

b $f(10)$

3 If $f(x) = -6x + 4$, find:

a $f(4)$

b $f(0)$

4 If $f(x) = 4x + 4$, find:

a $f(2)$

b $f(-5)$

5 If $f(x) = 3x - 1$, find:

a $f(3)$

b $f(-4)$

6 If $g(x) = \dfrac{7x}{4}$, find:

a $g(5)$

b $g(-4)$

7 Consider the function $f(x) = 4 + x^3$.

a Evaluate $f(4)$

b Evaluate $f(-2)$

8 Consider the function $f(x) = 2x^2 - 2x + 5$. Evaluate $f\left(\dfrac{1}{2}\right)$.

9 Consider the function $f(x) = \sqrt{5x + 9}$. Find the exact value of:

a $f(0)$

b $f(2)$

c $f(-1)$

10 If $f(t) = \dfrac{t^3 + 27}{t^2 + 9}$, evaluate the following:

a $f(-3)$

b $f(3)$

c $f(4)$

11 Consider the function $f(x) = x^2 + 8x$. Write an expression for:

12 Consider the function $f(x) = 2x^3 + 3x^2 - 4$.

a Evaluate $f(0)$.

b Evaluate $f\left(\frac{1}{4}\right)$.

13 If $j(x) = 3^x - 3^{-x}$, find the following, rounding your answers to two decimal places if necessary:

a $j(0)$

b $j(1)$

c $j(4)$

14 If $m(x) = \sqrt{12^2 - x^2}$, find:

a $m(4)$

b $m(0)$

c $m(9)$

d $m(\sqrt{2})$

15 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$.

b Form an expression for $p(m)$.

16 Consider the equation $x + 3y = 6$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$ for y.

c Find the value of $f(3)$.

17 Consider the equation $x - 4y = 8$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f : x \longrightarrow$.

c Find the value of $f : 12$.

18 Consider the equation $y + 6x^2 = 3 - x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(3)$.

19 Consider the equation $y - 4x^2 = 5 + x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f : x \longrightarrow$.

20 Consider the equation $-6x + 5y = 7$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f : x \longrightarrow$.

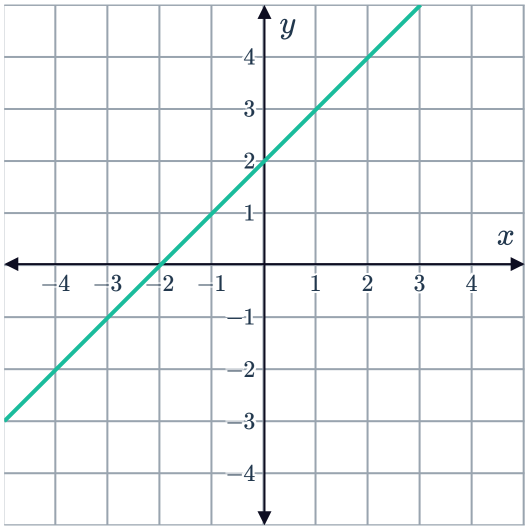
c Find the value of $f : 3$.

21 Use the graph of the function $f\left(x\right)$ to find each of the following values:

a $f\left(0\right)$

b $f\left(-2\right)$

c The value of x such that $f\left(x\right) = 3$



22 Consider the relation $h : x \longrightarrow -x^2 + 6x - 6$.

a For any input value of x, state the maximum number of distinct output values h can produce.

b Is h a function? Explain your answer.

23 For each of the following pairs of variables, determine which is the independent variable and which is the dependent variable:

a Cost of pizza and size of pizza.

b Mark achieved on a test and time spent studying.

c Duration of a loan and amound borrowed.

d Time spent exercising and physical fitness level.

24 Consider the function $f : x \longrightarrow x^2 - 49$.

a Find:

i $f(1)$ ii $f(8)$ iii $f(0)$

b If $f : x \longrightarrow 12$, what are the possible values for x, rounded to two decimal places?

25 Consider the function $g(x) = ax^3 - 3x +$ [redacted]

a Form an expression for $g(k)$.

b Form an expression for $g(-k)$.

c Is $g(k) = g(-k)$?

d Is $g(k) = -g(-k)$?

Applications

26 If $Z(y) = y^2 + 12y + 32$, find y when $Z(y) = -3$.

27 A function f is defined by $f : x \longrightarrow (x+4)\left(x^2 - 4\right)$.

a Evaluate $f(6)$.

b Find all solutions for which $f : x \longrightarrow 0$.

28 A graph of a quadratic equation of the form $y = ax^2 + bx + c$ passes through the points $(0, -7)$, $(-1, -8)$ and $(4, 77)$.

a Using the point $(0, -7)$, find the value of c.

b Substitute c and $(-1, -8)$ into the equation $y = ax^2 + bx + c$ to obtain an equation that describes a in terms of b.

c Substitute c and $(4, 77)$ into the equation $y = ax^2 + bx + c$ to obtain a simplified second equation in terms of a and b.

d Solve for a and b.

29 The financial team at The Gamgee Cooperative wants to calculate the profit, $P(x)$, generated by producing x units of wetsuits.

The revenue produced by the product is given by the equation is $R(x) = -\dfrac{x^2}{4} + 40x$. The cost of production is given by the equation $C(x) = 5x + 410$.

The profit is calculated as $P(x) = R(x) - C(x)$.

a Find an expression for $P(x)$ in terms of x.

b Find the values of the following:

i $R(70)$

ii $C(70)$

iii $P(70)$

c Sketch the graphs of $y = R(x)$, $y = C(x)$ and $y = P(x)$.

30 Elizabeth wants to calculate the cost to tra[illegible] Tehran and then Mexico City at certain times of the year. A program on her computer calculates $T(x)$, the total cost of this trip, by adding the cost of travelling to Tehran, $S(x)$, and the cost of travelling to Mexico City, $U(x)$.

Based on historical data, the computer program uses the calculations:

$$S(x) = x^2 - 200x + 10\,227 \quad \text{and} \quad U(x) = 18x + 15\,423$$

where x is the number of days until the date of travel.

a Find an equation for $T(x)$.

b Find the values of the following:

i $S(91)$ **ii** $U(91)$ **iii** $T(91)$

c Sketch the graphs of $y = S(x)$, $y = U(x)$ and $y = T(x)$.